FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	593561
Award Program	2018 New Innovator Award
Project Title	A coupled natural-human system approach to solving locust plagues
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Grant End Date	02/28/2022
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Total FFAR Budget	$298,834.79
Matching Funder(s)	$308,894.69
Final Report Date	05/27/2022
Project Director/Principal Investigator Information	
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General Information

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

- Tempe, Arizona, Congressional District 9, AZ-009
- Columbia Falls, Montana, Congressional District 1 (At-Large), MT-001
- Kaffrine, Senegal
- Dubbo, Australia
- Canberra, Australia
- Tucuman, Argentina
- Montpellier, France
- Amsterdam, Netherlands

1.2. How many new jobs were created because of this grant funding?

Year 1 = 2 jobs (Clara, Mira)
Year 2 = 3 (Mira, Geoffrey, Emmaline)
Year 3 = 2 (Sydney, Hannah)
Cumulative new positions created = 7

1.3. How many jobs were maintained because of this grant funding?

Year 1 = 5 (Arianne, Andrea, Douglas, Jonah, Marion)
Year 2 = 7 (Arianne, Andrea, Douglas, Jonah, Marion, Marty)
Year 3 = 6 (Arianne, Andrea, Geoffrey, Mira, Marion)
Cumulative unique positions maintained through support of this project= 8

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Locusts and grasshoppers continue to have devastating impacts on agriculture and food security around the globe. This was underscored by multiple outbreaks across 2019-2022 in East Africa, Southwest Asia, Southern Africa, and North and

South America, causing substantial agricultural losses and millions of hectares to be treated with pesticide. We advanced three primary objectives, which forged innovative management pathways and will provide benefits to any stakeholders affected by locust and grasshopper outbreaks and management, including the US food system.

(1) Research connections between land use practices and locusts and grasshoppers. Working side-by-side with partners, we carried out fieldwork in Senegal, Australia, and Argentina, as well as lab-based research at Arizona State University, US. Key findings included advancements in locust biology:

1. Corroborating earlier research on a few locust species, all new species studied preferred and performed best on plants with low protein and high carbohydrate contents. These "donut diets" support energetically costly migration.
2. It was previously hypothesized that field populations might shift their nutrient preferences to match what was available in their environment through space and time. However, we showed that locusts tend to maintain a consistent nutrient preference across space and time. Adult locusts likely obtain the appropriate balance of nutrients they need by migrating across the landscape, but populations of juvenile locusts (that cannot fly) are more susceptible to local nutrient limitations.
3. Because most locusts live in arid environments, rainfall that promotes vegetation growth can lead to outbreaks because it gives locusts plenty to eat. However, the effect of preceding vegetation on outbreaks varied significantly by geographic zone, bioregion, and season. By accounting for this hierarchical structure, locust forecasting models can greatly improve their capacity to predict outbreaks in response to rainfall.

(2) Compare locust governance structures and identify barriers for implementation of sustainable practices. Locusts are a peculiar case of large-scale extreme events that, despite being considered one of the most destructive migratory pests in the world by the United Nations, has been poorly studied by social sciences and governance studies. This objective supported cross-sectoral stakeholder workshops in South America and Australia, which were the first of their kind, and laid the foundation for future social science locust research. In the context of global change that imposes tremendous uncertainties and risks for the food system, this project supported locusts to become a textbook case that can inform collaborative governance for managing and responding to large-scale disasters.

(3) Link stakeholders to increase global capacity. Despite their global impact, limited groups work on locusts anywhere in the world and they are particularly sparse in the global south where communities are most vulnerable to locust outbreaks. This objective supported development of the Global Locust Initiative network that promotes opportunities for global stakeholders to network and collaborate, and now includes members from 42 countries.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

See section 2.3

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Objective 1: Research connections between land use practices and locusts and grasshoppers

- Understanding the connections between land use and grasshopper and locust outbreaks provides an alternative management strategy to chemical pesticides for land managers. This research was done in partnership with farmers and plant protection organizations, and thus directly informs management, including land management practices as well as forecasting models. For example, results from our collaborative work on Australian plague locusts is being incorporated to improve forecasting models led by the Australian Plague Locust Commission. In addition, through our partnership with the USDA APHIS Rangeland Grasshopper and Mormon Cricket team, they are incorporating findings on grasshopper nutritional preference supported by this FFAR project into optimizing grasshopper baits for the western US.

Objective 2: Compare locust governance structures and identify barriers for implementation of sustainable practices

- There are clear insights and benefits for anyone specifically governing locusts or migratory grasshoppers. For example, participants of the South American and Australian workshops hosted by this project immediately took insights gained from conversations and syntheses of the workshops into consideration for their own operations. As we share the results of the comparative locust governance study more broadly, more regions can benefit from these comparative insights, including the US National

Grasshopper Management Board, many members of whom are active in our global network (Objective 3).
- Beyond locusts and grasshoppers, governance insights from this project could be applied more broadly to other transboundary migratory pests and food systems challenges.

Objective 3: Link stakeholders to increase global capacity
- Despite their global impact, limited groups work on locusts anywhere in the world and they are particularly sparse in the global south where communities are most vulnerable to locust outbreaks. Prior to the Global Locust Initiative at Arizona State University (Founding Director is PI Arianne Cease) launching in 2018, there was no global organization to link partners and knowledge. This FFAR award provided critical support for us to grow the initiative, to develop tools for the network, and promote and support opportunities for global stakeholders to network and collaborate. For example, tools, such as our online community platform "HopperLink", created by this project mean that anyone in any region responding to an outbreak can immediately connect with the global experts on the topic in any language that google translates.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

See section 2.6 where all objectives, major activities, and significant results are summarized in aggregate.

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Roughly 2/3 of this project was carried out during the global pandemic. In response to the exceptional conditions imposed by COVID-19, we pivoted across all three objectives and still met our major goals for this award.

Objective 1: We adjusted the timing of some biology research (Obj. 1) to best align with logistical challenges, including timing of outbreaks, student schedules, and COVID. However, we were able to complete all major milestones.

Objective 2: The timing of this objective was most impacted by COVID because it involved large in person stakeholder workshops. We completed the South America regional workshop in person in February 2020, we shifted the Australia regional workshop to be virtual (March 2021), and were unable to complete the originallyplanned workshop based in Senegal, West Africa due to COVID travel restrictions and the technical challenges of poor internet access by the stakeholder communities. However, we replaced the focus on the governance of the Senegalese grasshopper with the desert locust. Because the desert locust has received the most attention of any locust, there was already a lot of published information to draw from. The 2019-2022 desert locust upsurge also brought much attention to this species and its governance, which further supported our work. Nevertheless, we were able to complete all major aims for this objective by shifting the locust species focus and adding complementary methodologies such as 1) a media-content analysis to explore the media dissemination of information on these outbreaks, how likely they are to contribute to shaping public perceptions of locusts, and who is in charge of their control and how to manage them sustainably; 2) an additional round of interviews to analyze governance in a context of crisis, specifically focusing on the current desert locust crisis in Eastern Africa.

While the postdoctoral researcher leading Objective 2 (Clara Therville) had to delay some project work due to maternity leave, she was recruited as a permanent researcher to the French National Research Institute for Sustainable Development. She included the governance of locust plagues as part of her research program and thus will continue to work on projects building on this FFAR-supported award into the future.

Objective 3: We adjusted some of the specific aims of this objective in response to iterative communication with global stakeholders and to overcome COVID restrictions. For example, in place of an in-person workshop, we developed an online global community where anyone with access to the internet can connect and share ideas directly in any language that google translates.

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Objective 1: Research connections between land use practices and locusts and grasshoppers (100% complete)

Working side-by-side with partners, we carried out fieldwork in Senegal, Australia, and Argentina, as well as lab-based research at Arizona State University (ASU). This involved multiple field trips to each location and three years of near continuous lab work at ASU. Here are some key findings for the three focal species:

<u>Species #1: Australian plague locust (*Chortoicetes terminifera*)</u> is distributed

throughout arid and semi-arid Australia. It is a major agricultural pest and managed at a national level by the Australian Plague Locust Commission (APLC).

- Work led and described by former PhD student Douglas Lawton:
    1. My research contributes to FFAR objective 1 by increasing our understanding of preceding indicators for Australian plague locust swarms. This is done on multi-scales through field, lab, and remote sensing approaches. The research can be broken into multiple parts:
        1. Australian plague locusts need to acquire the optimal amount of nutrients within a variable nutritional landscape. The Australian plague locust maintained a consistent self-selected protein to carbohydrate ratio regardless of what was available within local environments. This suggests that this species must migrate or post-ingestively regulate to meet physiological demands. A strong metapopulation structure may aid in the persistence of locusts at larger spatial scales by allowing them to shift among different habitats to find suitable nutritional environments. This manuscript was published in the journal Ecosphere in 2021.
        2. Understanding what landscapes will induce gregarization and subsequent locust band marching is important for correctly identifying these landscapes in survey protocol. I was interested in seeing how the distribution of nutritionally optimal and suboptimal food would induce gregarization of Australian plague locust nymphs. I showed that a clumpy distribution of nutritionally optimal food influences where locusts eat within the landscape and this can be translated into individual gregarization. Research is needed to further support my findings but thinking about how nutritionally optimal and suboptimal food sources are distributed within a landscape is likely important for the gregarization of nymphs which may lead to band formation and subsequent swarming.
        3. The ecological causes of locust swarms happen on different spatiotemporal levels. For example, gregarization happens to individual locusts, which have population to landscape level effects. For this research, I investigated how the relationship between locust outbreaks and preceding vegetation varied spatiotemporally. I show the relationship between preceding vegetation and locust outbreaks varied significantly between bioregions and seasonally. However, the structure of geographic zone > bioregion > season held constant. This spatiotemporal hierarchy was done in two locust species. Therefore, the results here are likely generalizable for many locust species. Findings of this study are being implemented into the building of the new Australian plague locust forecasting model. This manuscript was published in the journal Ecography in 2022.
- Former MS student Jonah Broseman and undergraduate honors student

Sydney Millerwise led lab-based research testing cereal crop preference of Australian plague locusts.

1. Sydney's thesis showed that plant toughness was an important factor determining the preference of the Australian plague locust for different crop species
2. Jonah is in the final stages of manuscript preparation prior to submitting to a peer-reviewed journal. Here is his abstract:
    1. In contrast to predictions from nitrogen limitation theory, recent studies have shown that herbivorous migratory insects tend to be carbohydrate (not protein) limited, likely due to increased energy demands, leading them to preferentially feed on high carbohydrate plants. However, additional factors such as mechanical and chemical defenses can also influence host plant choice and nutrient accessibility. In this study, we investigated the effects of plant protein and carbohydrate availability on plant selection and performance for a migratory generalist herbivore, the Australian plague locust, *Chortoicetes terminifera.* We manipulated the protein and carbohydrate content of seedling wheat (*Triticum aestivum L.*) by two means: 1) we increased the protein:carbohydrate ratio using fertilizer (N-P-K 46-0-0), 2) we sought to increase carbohydrate accessibility by grinding cell walls after drying the plants. Using a factorial design, we ran both choice and no-choice experiments to measure preference and performance. We confirmed locust preference for plants with a lower protein-carbohydrate ratio (unfertilized plants). Unlike previous studies with mature wild grass species, we found that intact plants supported better performance than dried and ground plants, suggesting that cell wall removal may only improve performance for tougher or more carbohydrate-rich plants. These results add to the growing body of evidence suggesting that several migratory herbivorous species perform better on plants with a lower protein:carbohydrate ratio.

<u>Species #2: Senegalese grasshopper (*Oedaleus senegalensis*)</u> is a major pest across the Sahel, predominantly attacking staple cereal crops including millet, sorghum, and maize. In Senegal, it is managed at a national level by the Senegalese Plant Protection Directorate (DPV).

- Work on this species was led by postdoctoral researcher Marion Le Gall and built on earlier work where we showed that Senegalese grasshoppers preferred and performed best on low protein grasses found in degraded fields.
    1. She published two papers with support from this award, and is working on a third. Here are the abstracts that contextualize this research and highlight the major advancements:

1. There is generally a close relationship between a consumer's food and its optimal nutrients. When there is a mismatch, it is hypothesized that mobile herbivores switch between food items to balance nutrients, however, there are limited data for field populations. In this study, we measured ambient plant nutrient content at two time points and contrasted our results with the nutrient ratio selected by wild female and male grasshoppers (Oedaleus senegalensis). Few plants were near O. senegalensis' optimal protein:carbohydrate ratio (P:C), nor were plants complementary. Grasshoppers collected earlier all regulated for a carbohydrate-biased ratio but females ate slightly more protein. We hypothesized that the long migration undertaken by this species may explain its carbohydrate needs. In contrast to most laboratory studies, grasshoppers collected later did not tightly regulate their P:C. These results suggest that field populations are not shifting their P:C to match seasonal plant nutrient shifts and that mobile herbivores rely on post-ingestive mechanisms in the face of environmental variation. Because this is among the first studies to examine the relationship between ambient nutrient landscape and physiological state our data are a key step in bridging knowledge acquired from lab studies to hypotheses regarding the role ecological factors play in foraging strategies. This was published in Current Research in Insect Science in 2021.
2. Multivoltine insects can produce multiple generations in one year. Favorable conditions support more generations, leading to serious outbreaks. For herbivores, plant nutrient availability is a major environmental factor affecting fitness and it can shift substantially throughout seasons. In a stochastic environment, organisms can adopt several strategies to regulate their nutrient intake and maximize performance. However, data regarding nutrient regulation of wild herbivores are scarce, and even more so regarding potential intergenerational plasticity. To bridge this gap, we measured nutritional regulation and performance of an outbreaking multivoltine herbivore – one of the most serious agricultural pests in the Sahel: Oedaleus senegalensis. We surveyed a field population in Senegal and measured its nutritional preference and regulation across two generations (G1 and G3) using artificial diets and plant choice experiments. In the field, G1 locusts were five to ten times more abundant than G3 locusts. We found that G1 and G3 locusts selected different protein:carbohydrate ratios but also that the strength of regulation was different. G1 locusts regulated their nutrient target more tightly than G3 locusts. In contrast, studies with laboratory populations demonstrate

strong regulation for grasshoppers, appearing less plastic than field populations. Both generations selected a carbohydrate-biased nutrient ratio, although it was more carbohydrate-biased for G3 locusts. In both cases, plant nutrient contents in the field were more protein biased than their preferred diet. Therefore, choices by locusts were likely influenced by other ecological variables such as leaf toughness or plant defenses. G1females were heavier and laid more eggs than G3 females. However, G3 locusts survived longer during the experiment than G1 locusts, suggesting a potential generational tradeoff between reproduction and survival. Our data highlight the importance of studying nutritional regulation in situ and incorporating field and lab data to better understand foraging decisions and nutritional tradeoffs. This was published in Oikos in 2022.

3. In a third project, Marion and partners compared the migratory patterns of the Senegalese locusts with the nutrient contents of the available plant communities at five villages across a south to north route in Senegal. To do this, Marion and the team worked with women's groups in the villages during Years 1 and 2 to collect and report locust activity. During Years 2 and 3 they completed lab analyses on the field samples at ASU.

Species #3: South American locust (*Schistocerca cancellata*) was a devastating pest in South America up until the 1950s, after which outbreaks became rare and only reported in a small region in Argentina. However, 2015 brought an upsurge on a scale not seen in 60 years, which expanded into Bolivia and Paraguay. Our ASU team made multiple trips to these three countries, working with stakeholders to study the nutritional ecology of outbreaks and providing global connections and resources to help re-build locust response infrastructure. This species is managed at a national level by SENASA and INTA (Argentina), SENAVE (Paraguay), and SENASAG (Bolivia).

- A major advancement supported by all three objectives of the FFAR project was the first comprehensive review of the biology, ecology, and management of this species in 60 years. The interdisciplinary and cross-sectoral review was led by INTA scientist, Eduardo Trumper, and included 12 authors from 9 different organizations. Here is the abstract:
    1. In the first half of the twentieth century, the South American Locust (SAL), *Schistocerca cancellata* (Serville, 1838), was a major pest of agriculture in Argentina, Bolivia, Paraguay, Uruguay, and Brazil. From 1954–2014, a preventive management program appeared to limit SAL populations, with only small- to moderate-scale treatments required, limited to outbreak areas in northwest Argentina. However, the lack of major locust outbreaks led to a gradual reduction in resources, and in 2015, the sudden appearance of swarms marked the beginning of a substantial upsurge, with many swarms reported

initially in Argentina in 2015, followed by expansion into neighboring countries over the next few years. The upsurge required a rapid allocation of resources for management of SAL and a detailed examination of the improvements needed for the successful management of this species. This paper provides a review of SAL biology, management history, and perspectives on navigating a plague period after a 60-year recession. This was published in a special issue of the journal Agronomy in 2022.

Objective 2: Compare locust governance structures and identify barriers for implementation of sustainable practices (100% complete)

Locusts are a peculiar case of large-scale extreme events that, despite being considered one of the most destructive migratory pests in the world by the United Nations, has been poorly studied by social sciences and governance studies. To increase capacity, PI Arianne Cease collaborated with ASU Professor Marty Anderies to recruit and co-advise postdoctoral scholar Clara Therville. Clara led studies on the connectivity and robustness of locust management across institutions, sectors, and continents in an effort to advance sustainable locust management. These studies included cross-sectoral stakeholder workshops in South America and Australia, which were the first of their kind. The team wrote a foundational paper to support additional social science research (Therville et al. 2021).

In Feb. 2022, Clara started her own research team as a permanent researcher at IRD (French National Research Institute for Sustainable Development) - UMR SENS - Montpellier, France, and is already advising masters student Hannah Korinth to do her thesis on desert locust governance. To our knowledge, Clara's will be the first social science and sustainable development team to have a substantial focus on locusts and other transboundary migratory pests, a milestone that would not have been possible without this FFAR award.

Specific outcomes:

- To illustrate the importance of social science to entomologists, and introduce social scientists to locusts, we published a scientific article on the role of social sciences for sustainable locust management. This paper published in the journal Agronomy (2021) aims to promote interdisciplinary approaches by examining the scope and purpose of different subfields of social science and explores how they can be applied to different issues faced by entomologists and practitioners to implement sustainable locust research and management. In particular, we discuss how environmental governance studies resonate with two major challenges faced by locust managers: implementing a preventative strategy over a large spatial scale and managing an intermittent outbreak dynamic characterized by periods of recession and absence of the threat.
- South America and Australian stakeholder workshops: We held a governance workshop in Argentina in 2020. The synthesis of this workshop

has been translated into Spanish and shared with all participants. This work mobilized a collaborative scientific publication by Trumper et al. in the journal Agronomy (see Objective 1 results). For the Australian case study, we held a virtual workshop, accompanied by an online survey in early 2021. The goals were similar as for the Argentinian workshop: explanation of the last outbreak situation, characterization of the governance structure, its strengths and weaknesses, and analysis of the future challenges. The results will be shared in a synthesis that will be released in 2022.

- Desert locust governance and response: As the COVID-19 situation did not allow us to work in Senegal, we choose to focus on the desert locust using two complementary approaches: 1) complementary virtual interviews to analyze the 2019-2021 desert locust outbreak and related governance challenges and 2) a media-content analysis to explore the media dissemination of information on these outbreaks and compare the 2003-2005 crisis with the 2019-2021 crisis and identify possibly shifting frames.
    1. Regarding the first approach, complementary interviews were conducted by Hannah Korinth, a research masters student at the University of Amsterdam in International Development, under the supervision of Clara Therville. Seventeen interviews have been conducted, and the collected material is currently being analyzed. Preliminary results demonstrate the active effort of actors and institutions to engage in collective action to manage the outbreak, despite initial coordination and logistical challenges. New developments were seen in the technology and tools used in the outbreak, which aided clearer communication and faster coordination. This case study however highlights a divergence of the perceived roles and responsibilities amongst the actors at all levels but, in particular, surrounding the discussion of the role of international stakeholders. Finally, the results indicate that large scale transboundary pests are effectively governed with centralized coordination and decentralized operation effort. In the long-term sustainable funding is required with a clear understanding amongst all stakeholders who will take the central coordination role for locust management. This work will be first released in the form of a master thesis, and was accepted as a poster communication at the International Geographic Union Congress, Paris 2022.
    2. Regarding the second approach, most data were collected by Maddie Magrino, an undergraduate student at Arizona State University supervised by Clara Therville. The analysis of the data has begun but needs to be completed, in partnership with Marie Chandelier, a linguist based at Montpellier University.
- Building from these three case-study comparisons, one scientific publication is in the process of being written: (Governing transboundary extreme events: lessons for locusts). We highlight that locust managers have implemented multi-level, nested, and adaptive systems, from local to international levels, more or less centralized depending on the stage and the context. While these systems necessarily have to present a high level of

adaptiveness to face the extended, uncertain, and discontinuous dynamic imposed by locusts, such adaptiveness results in blurring the distribution of responsibilities and questions actors' willingness and capacities to fulfill their roles through space and time - especially under a reduced frequency and amplitude regime of perturbation. We identify the responses implemented by locust managers around the world to face this variability and analyze their inherent fragilities. We discuss how global trends towards decentralization and inclusion of local levels, and increased consideration of environmental issues, albeit essential, have intensified variability in the represented values, interests, and action capacities of the actors. In addition to the inherent spatio-temporal dynamics, these societal shifts further challenge collaborative governance for managing and responding to large-scale disasters. We conclude that in a context of global change that imposes renewed conditions of interconnectedness, hazards, and vulnerabilities, locusts are a textbook case that can inform collaborative governance for managing and responding to large-scale disasters. This work has been accepted for an oral communication at the International Geographic Union Congress, Paris 2022.

Objective 2: Link stakeholders to increase global capacity (100% complete)

Despite their global impact, limited groups work on locusts anywhere in the world and they are particularly sparse in the global south where communities are most vulnerable to locust outbreaks. Prior to the Global Locust Initiative at Arizona State University (Founding Director is PI Arianne Cease) launching in 2018, there was no global organization to link partners and knowledge. This FFAR award provided critical support for us to grow the initiative, to develop tools for the network, and promote and support opportunities for global stakeholders to network and collaborate. Efforts for this objective were led by Global Locust Initiative Project Coordinator, Mira Word Ries, and GLI Co-Director Rick Overson. This objective provided the foundation for the first two objectives. Here are some highlights:

- Throughout the project, we responded in real time to our partners who were in the midst ongoing locust outbreaks to share information and resources. For example, Hector Medina, the head of SENASA Argentina's locust response requested information on aerial control and we were able to quickly connect him with Chris Adriaansen, director of the Australian Plague Locust Commission, who shared documentation of all their best practices for aerial control. We also responded to dozens of media requests and shared information about locust research, response, and management through a myriad of media venues.
- We hosted a panel event on database management at the International Congress of Orthopterology in Morocco in March 2019. Panel members included key representatives from six continents and the resulting discussion laid the foundation for the themes of the international workshop on locust monitoring and forecasting.
- In place of planned in-person activities (i.e., international workshop on

locust monitoring and forecasting), we created an online community platform for the Global Locust Initiative network, HopperLink. HopperLink has become a critical hub for global communication and real-time updates on locust management with 180+ participants from 42 countries including farmer's groups, students, researchers, and government officials. New members have been joining weekly since its launch in January 2021. This online community additionally allows for an efficient way to engage in meaningful outreach activities and to receive feedback from broad stakeholder groups. Project Coordinator Mira Word hosts the platform by facilitating new member onboarding, submitting weekly discussion posts, responding to questions, and featuring network member's work.

- To promote transdisciplinary work, Mira is coordinating the first comprehensive review of institutional perspectives on locust research and management. This manuscript is targeted for submission in 2022 and includes 40 co-authors from six continents with representatives from natural and social science, as well as government and international organizations. (see section 3.2)

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

This complex project spanned multiple natural and social science disciplines and therefore used multiple methodologies and ways of knowing. We empowered the leads of each objective and sub-project to develop and carry out research according to best practices of each disciplinary technique used, with consistent support and feedback from the broader team and outside experts in those fields.

2.8. List key stakeholders that could be served by results of this project.

Any stakeholders that are involved in or affected by locust and grasshopper management, or related activities, could be served by the results of this project. This includes any agriculturist whose crops or fields are susceptible to these pests. While this can include an extensive list of surprising crops (including citrus and tobacco!), locusts and grasshoppers who have massive outbreaks tend to target grazing pastures and cereal crops first. Local, regional, national, and international plant protection organizations are also a key stakeholder group, and we maintain strong connections and communications with these groups. Additional stakeholders range from other government and non-governmental organizations and businesses to ecologists, social scientists, and other academics.

For a list of key grasshopper and locust organizations: https://sustainability-innovation.asu.edu/global-locust-initiative/resources/

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Objective 1: Key stakeholders from plant protection organizations participated as partners and co-creators on the lab and field research and were co-authors on these publications. They further shared them with their communities. (See section 3.2 for a list of publications; all papers had at least one co-author from a government organization involved in plant protection). In addition, Douglas Lawton produced an overview of the research he led in Australia for the Australian Plague Locust Commission (APLC), which was highlighted within the APLC annual report for 2019-2020.

Objective 2: Clara Therville produced syntheses of the South American and Australian workshops, which were translated and shared with participants.

Objective 3: We disseminated information broadly through our website (locust.asu.edu) and through communications shared with the Global Locust Initiative network.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

See section 2.5

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

This project provided opportunities for 2 undergraduate students (Sydney Millerwise, Maddie Magrino), 2 Masters students (Jonah Brosemann, Hannah Korinth), 2 PhD students (Douglas Lawton, Emmaline Gates), and 2 postdoctoral researchers (Marion Le Gall, Clara Therville).

All mentees met regularly with their advisors to set project milestones, track progress, and ensure they are advancing their Individual Development Plans. Mentees participated in regular lab and project meetings, presenting their data to the research team, as well as at international conferences. Career development is a strong focus. All are well-poised to continue to be successful in their careers.

Sydney and Maddie both graduated and were accepted into masters programs. Jonah received his MS and now works for an environmental consultant company. Hannah received her masters and was accepted into a PhD program. Douglas received his PhD, was a postdoctoral scholar for a year at NC State, and just accepted a permanent position as a data scientist for AgBiome. Marion Le Gall was recruited as an Instructor at Arizona State University and works with honors students to connect them with research opportunities. Clara Therville was recruited as a permanent researcher at IRD - Research Institute for Development - in Montpellier, France.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Across all years, a total of 8 students and postdocs were directly funded by and engaged in leading different aspects of this project:

- Undergraduate students - 2 (Sydney Millerwise, Maddie Magrino)
- Graduate students - 4 (Jonah Brosemann, Douglas Lawton, Emmaline Gates, Hannah Korinth)
- Postdoctoral scholars - 2 (Marion Le Gall, Clara Therville)

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

See 3.2

3.2. Please provide a list of citations for the information products produced during the project.

Peer-reviewed publications:

1. Lawton D J, M Le Gall, C Waters and A. J. Cease (2021) Mismatched diets: defining the nutritional landscape of grasshopper communities in a

variable environment. Ecosphere 12(3):e03409. 10.1002/ecs2.3409

   a. Supporting information:
      - https://esajournals.onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1002%2Fecs2.3409&file=ecs23409-sup-0001-AppendixS1.pdf

2. Lawton D, Scarth P, Deveson E, Piou C, Spessa A, Waters C, Cease AJ (2022) Seeing the locust in the swarm: accounting for spatiotemporal hierarchy improves ecological models of insect populations. Ecography. https://doi.org/10.1111/ecog.05763
   a. Supporting information:
      - https://onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1111%2Fecog.05763&file=ecog12844-sup-0001-AppendixS1.pdf
      - https://onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1111%2Fecog.05763&file=ecog12844-sup-0002-AppendixS2.pdf

3. Le Gall M, Beye A, Diallo M, Cease AJ. (2022). Generational Variation in Nutrient Regulation for an Outbreaking Herbivore. *Oikos* n/a, no. n/a (April 28, 2022): e09096. https://doi.org/10.1111/oik.09096.
   a. Supporting information:
      - https://onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1111%2Foik.09096&file=oik13374-sup-0001-AppendixS1.docx
      - Data are available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.v15dv41xp> (Le Gall et al. 2022).

4. Le Gall M, Word ML, Beye A, Cease A. (2021) Physiological Status Is a Stronger Predictor of Nutrient Selection than Ambient Plant Nutrient Content for a Wild Herbivore. Current Research in Insect Science 1 (January 1, 2021): 100004. https://doi.org/10.1016/j.cris.2020.100004.
   a. Supporting information:
      - Data are available from the Dryad Digital Repository: https://datadryad.org/stash/dataset/doi:10.5061%2Fdryad.5dv41ns3w

5. Therville C, Anderies JM, Lecoq M, Cease A (2021) Locusts and People: Integrating the Social Sciences in Sustainable Locust Management. Agronomy 11: 951. https://doi.org/10.3390/agronomy11050951
   a. Supporting information:
      - https://www.mdpi.com/2073-4395/11/5/951/s1?version=1620782321

6. Trumper EV, Cease AJ, Cigliano MM, Bazán FC, Lange CE, Medina HE, Overson RP, Therville C, Pocco ME, Piou C, Zagaglia G, Hunter D (2022) A Review of the Biology, Ecology, and Management of the South American Locust, Schistocerca cancellata (Serville, 1838), and Future Prospects. Agronomy 12: 135. https://doi.org/10.3390/agronomy12010135

Manuscripts in preparation:

7. Brosemann J, Overson R, Cease A, Millerwise S, Le Gall M. Nutrient supply and accessibility in plants: effect of protein and carbohydrates on Australian plague locust (*Chortoicetes terminifera)* preference and performance
8. Mira Word [1], Chris Adriaansen[2], Kevin Berry[3], David Branson[4], Amadou Bocar Bal[5], Annie Byrne[6], Maria Cecilia Catenaccio[7], Maria Marta Cigliano[8], Darron Cullen[9], Ted Deveson[10, 11], Aliou Diongue[12], Bert Foquet[13], Joleen Hadrich[14], David Hunter[15], Dan Johnson[16], Juan Pablo Karnatz[17], Carlos Lange[18], Douglas Lawton[1], Mohammed Lazar[19], Alex Latchininsky[20], Michel Lecoq[21, 26], Marion Le Gall[1], Jeffrey Lockwood[22], Idrissa Maiga[23], Balanding Manneh[24], Rick Overson[1], Britt Peterson[25], Cyril Piou[26], Mario A Poot-Pech[27], Brian Robinson[28], Stephen Rogers[24], Hojun Song[13], Clara Therville[1], Eduardo Trumper[29], Cathy Waters[30], Derek A Woller[31], Long Zhang[32], Arianne Cease[1]. Global perspectives and approaches to locust and grasshopper management and research
9. Clara Therville[1], John Marty Anderies[1,3,4], Chris Adriaansen, Ted Deveson, Douglas Lawton[1], Hector E. Medina, Rick Overson[1], Eduardo V. Trumper, Arianne Cease[1,2]. Governing transboundary extreme events: lessons from locusts

Public news posts:

- The Global Locust Initiative has a website that provides details about the GLI, including its projects. The home page to the GLI website is below: https://sustainability.asu.edu/global-locust-initiative/
- As part of Objective 2, we hosted stakeholder workshops in Argentina and Australia. For summaries, see news post here:
    - https://sustainability-innovation.asu.edu/global-locust-initiative/news/archive/ffar-project-first-workshop-on-institutional-barriers-to-sustainable-locust-management-in-south-america/
    - https://sustainability-innovation.asu.edu/global-locust-initiative/news/archive/global-locust-initiative-hosts-second-stakeholder-workshop-on-locust-and-grasshopper-management-australia-edition/
- As part of Objective 2, we completed a media content analysis of the desert locust upsurges in 2004 and 2020. For a summary, see news post here: https://sustainability-innovation.asu.edu/global-locust-initiative/news/archive/comparing-media-coverage-of-desert-locust-outbreaks/
- As part of Objective 3, we hosted a panel event on database management at the International Congress of Orthopterology in Morocco in March 2019. Panel members included representatives from six continents and the resulting discussion laid the foundation for the themes of the international workshop on locust monitoring and forecasting. For a summary, see news post here: https://sustainability-innovation.asu.edu/global-locust-initiative/news/archive/global-locust-initiatives-post-review-of-the-13th-

international-congress-of-orthopterology/

- As part of Objective 3, we created an online professional network for the global locust and grasshopper community. Here is a news post when our community reached 100 members (since then, it has nearly doubled): https://sustainability-innovation.asu.edu/global-locust-initiative/news/archive/new-online-community-for-the-global-locust-network-reaches-100-members/

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Yes -- please see section 3.2

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Please see section 3.2

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Confidential information includes: audio records of the workshop in Argentina, audio of interviews, and interview transcripts. These are confidential because they provide personal information and will only be used for research purposes, according to our approved IRB protocol.

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

- Created blog posts summarizing our Argentina and Australian workshops and posted links on Twitter to these blog posts (see 3.2 for links).
- Shared the Argentina and Australia workshop summaries directly with participants and stakeholders
- Shared updates about this FFAR project with the Global Locust Network.

- Created a private, professional networking platform for the Global Locust Network to connect and share ideas: https://sustainability-innovation.asu.edu/global-locust-initiative/network/. To join: https://global-locust-network.mobilize.io/registrations/groups/43483

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

Objective 1: Lab and field research produced new empirical data on locust and grasshopper biology. These datasets are being made publicly available when manuscripts containing these data are published (see 3.2)

Objective 2: Stakeholder interviews, online surveys, and data collected from participants during the South American and Australian locusts produced confidential data. Anonymized analyses and syntheses of these data will be made publicly available when manuscripts containing these data are published (see 3.2 and 3.6).

Objective 3: All information curated through objective 3 is publicly available on our website locust.asu.edu and global community platform https://sustainability-innovation.asu.edu/global-locust-initiative/network/

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Not Applicable

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Not Applicable

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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: __Dr. Arianne Cease_________________________________

PI Signature: _____ ___________ Date: ___15 May 2022_______

Authorized Signing Official (ASO) Name: _______Mikayla Gerdes____________________

ASO Signature: Mikayla Gerdes Digitally signed by Mikayla Gerdes Date: 2022.07.22 15:29:26 -07'00' ______________________________ Date: ____________________